Generated from docs/ac30_schematica.jpg (SWTPC AC−30 Cassette Tape Switching Schematic),
docs/AC30−BOM.md and docs/ac30_wiring1a.jpg. Each block is wired; net labels join the blocks and sheets
(identical labels are connected).
Transcription notes / assumptions:

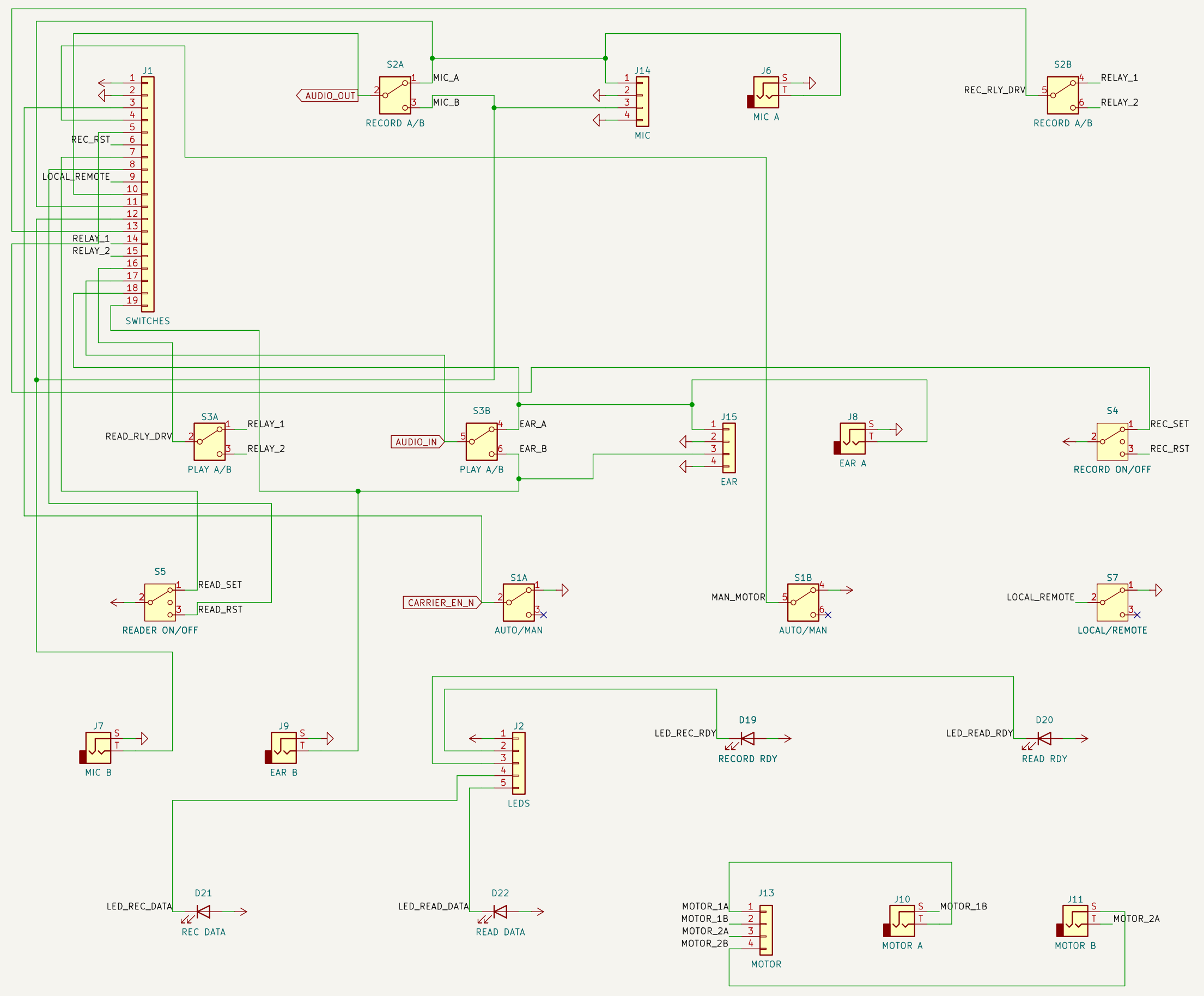
- Drawing pin−number slips corrected to the real pinouts: IC7D drawn 8−>9 (4049: 9−>10); IC13C drawn inputs 7,8 (4070: 8,9).
- 4053 INH (pin 6) and VEE (pin 7) are not shown on the drawing; tied to GND.
- IC10 pins 4/8 tie to the R38 supply node, assumed +5V.
- IC15 (1488) supply pins are not shown; taken as +13V / −13V from the power supply (sheet 3).
- CONTROL INVERT: drawn as R42 pull−up; it is also brought out on J4.
- J3−J5 pin positions match the original artwork (docs/ac30_redblue150.jpg).
- The front panel is wired to five rows of 0.1 in through−hole pads instead of the artwork's J1/J2 harness headers: J1 SWITCHES, J2 LEDS, J13 MOTOR, J14 MIC (MIC A/B jacks) and J15 EAR (EAR A/B jacks). The S2/S3 audio poles wire to the SWITCHES pads (AUDIO_OUT, MIC_A/B, AUDIO_IN, EAR_A/B); the board joins them.
- MIC/EAR jack reference designators J6−J9 are assigned arbitrarily; J10/J11 are the motor jacks.
- Trimmer R39 (DELAY, next to IC10 on the artwork) is not on any schematic; it is listed, not placed, in tools/netlist.py.
- The artwork's J2 pins 11/12 (18 VAC A/B) are gone: the power supply is external.

SMT variant (kicad−smt/AC30_SMT): every on−board part is surface mount except the Molex KK−396 connectors (J3−J5, J12), the front−panel jumper pads and the mounting holes. 0805 R/C, SOD−123 diodes (1N4148W, BZT52C4V7), SOT−23 transistors (MMBT5088 for 2N5210, MMBT5087), SOIC ICs, tantalum EIA−3216 (C4, C17), 6.3x7.7 mm aluminium C16, Bourns 3314G trimmer R16, 1206 LED D18. RLY1/RLY2 are Omron G6K−2F−Y (5 V coil, DPDT 1 A, both poles in parallel as the motor contact) instead of 6 V SIL reed relays; check the coil polarity mark (pin 1 to RELAY_x).

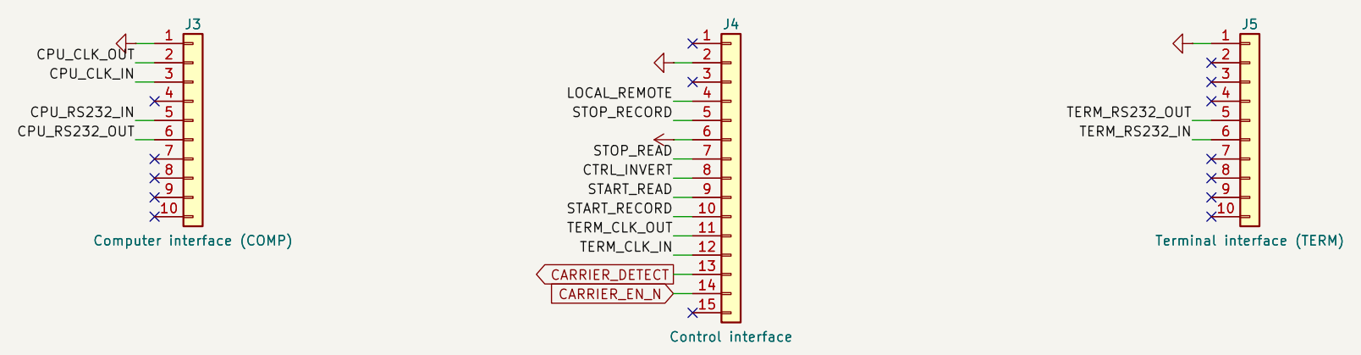
Mod/Demod & power input
File: AC30_SMT_moddemod_power.kicad_sch

Power supply (external)
File: AC30_SMT_power_supply.kicad_sch

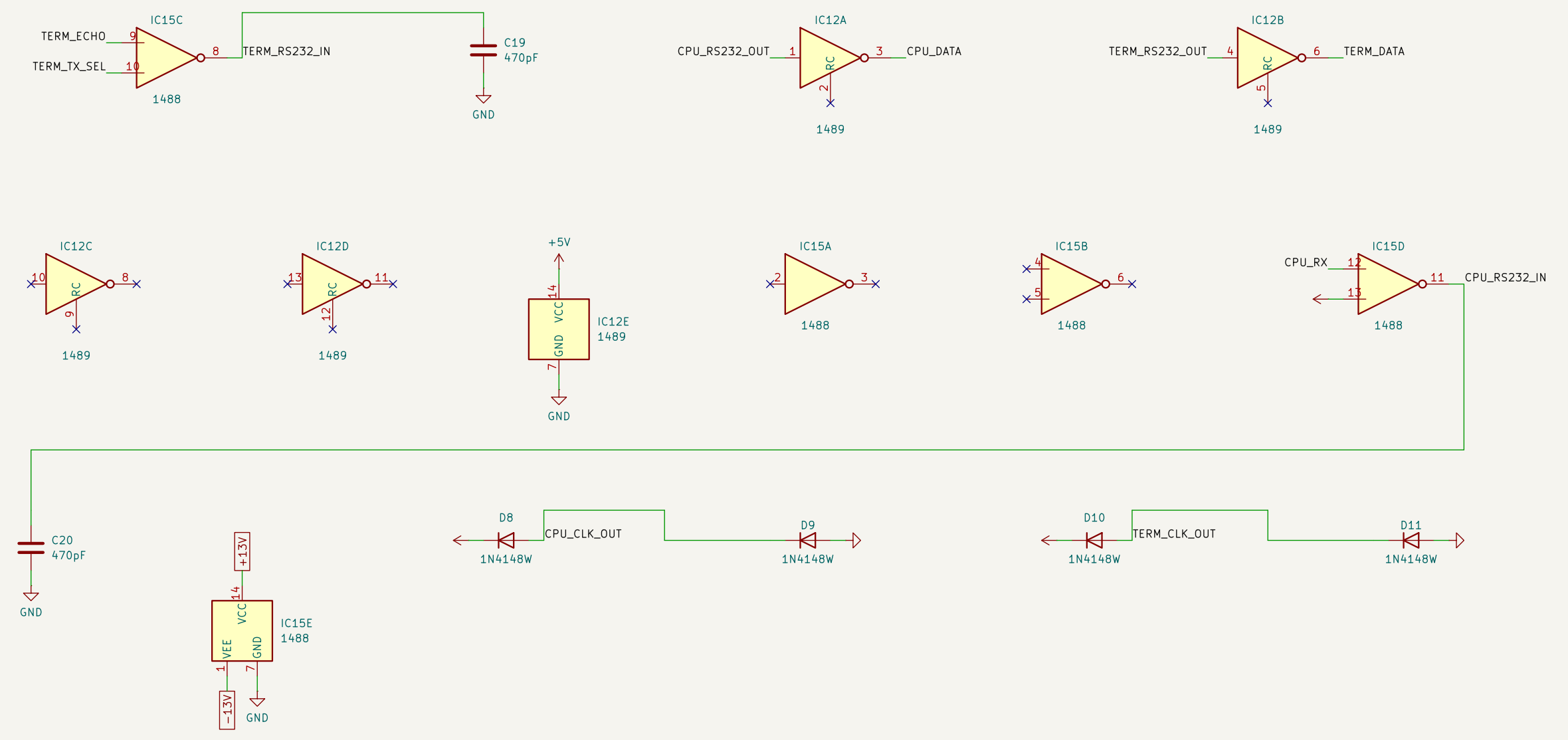
Front panel switches, LEDs and jacks


J1
SWITCHES
REC_RST
LOCAL_REMOTE
RELAY_1
RELAY_2
S2A
AUDIO_OUT
MIC_A
MIC_B
RECORD A/B
J14
MIC
J6
MIC A
S2B
REC_RLY_DRV
RELAY_1
RELAY_2
RECORD A/B
S3A
READ_RLY_DRV
RELAY_1
RELAY_2
PLAY A/B
S3B
AUDIO_IN
EAR_A
EAR_B
PLAY A/B
J15
EAR
J8
EAR A
S4
REC_SET
REC_RST
RECORD ON/OFF
S5
READ_SET
READ_RST
READER ON/OFF
S1A
CARRIER_EN_N
AUTO/MAN
S1B
MAN_MOTOR
AUTO/MAN
S7
LOCAL_REMOTE
LOCAL/REMOTE
J7
MIC B
J9
EAR B
J2
LEDS
D19
LED_REC_RDY
RECORD RDY
D20
LED_READ_RDY
READ RDY
D21
LED_REC_DATA
REC DATA
D22
LED_READ_DATA
READ DATA
J13
MOTOR_1A
MOTOR_1B
MOTOR_2A
MOTOR_2B
MOTOR
J10
MOTOR A
J11
MOTOR B

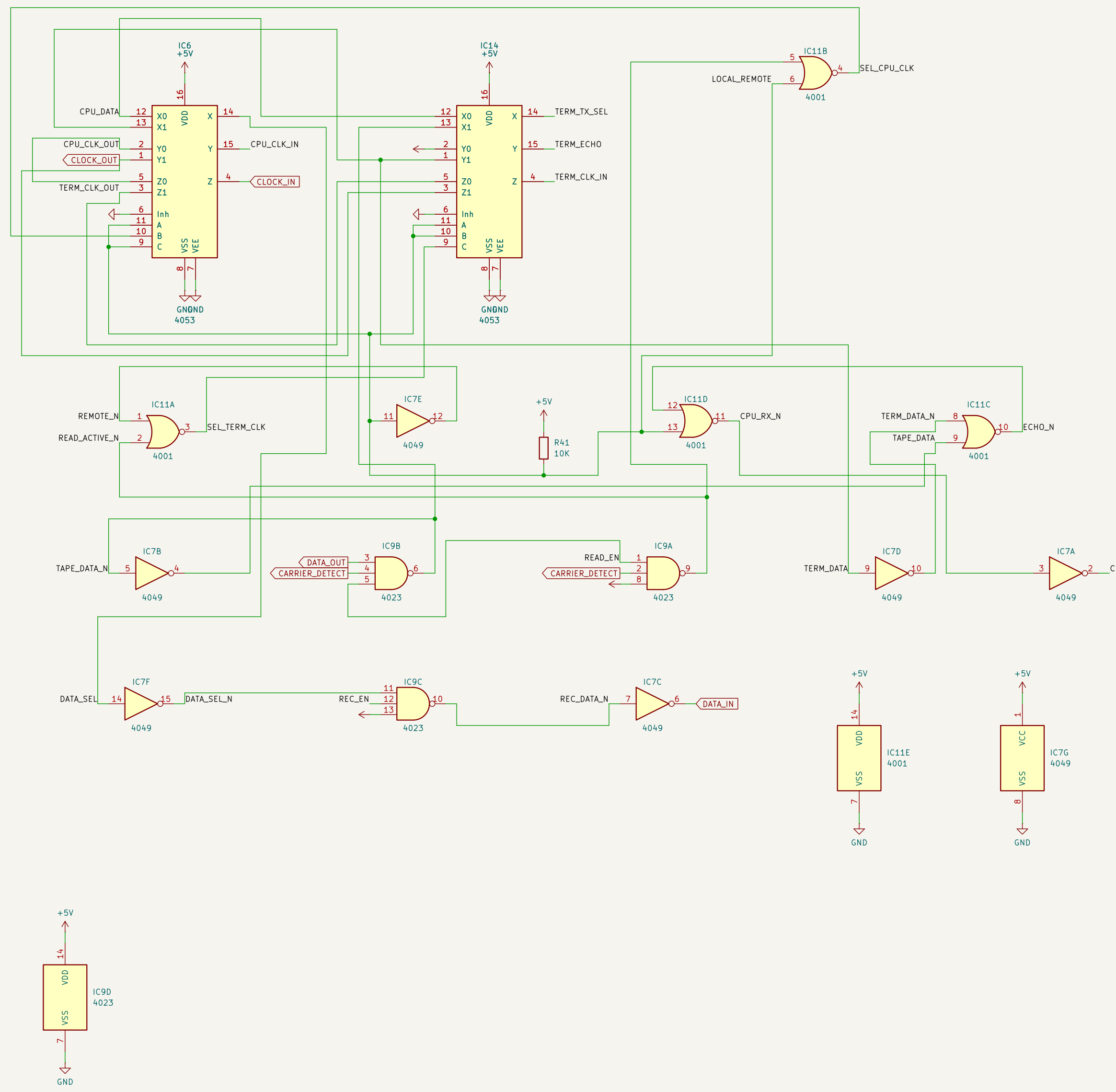
Board connectors (pin positions per original artwork)


J3
CPU_CLK_OUT
CPU_CLK_IN
CPU_RS232_IN
CPU_RS232_OUT
Computer interface (COMP)
J4
LOCAL_REMOTE
STOP_RECORD
STOP_READ
CTRL_INVERT
START_READ
START_RECORD
TERM_CLK_OUT
TERM_CLK_IN
CARRIER_DETECT
CARRIER_EN_N
Control interface
J5
TERM_RS232_OUT
TERM_RS232_IN
Terminal interface (TERM)

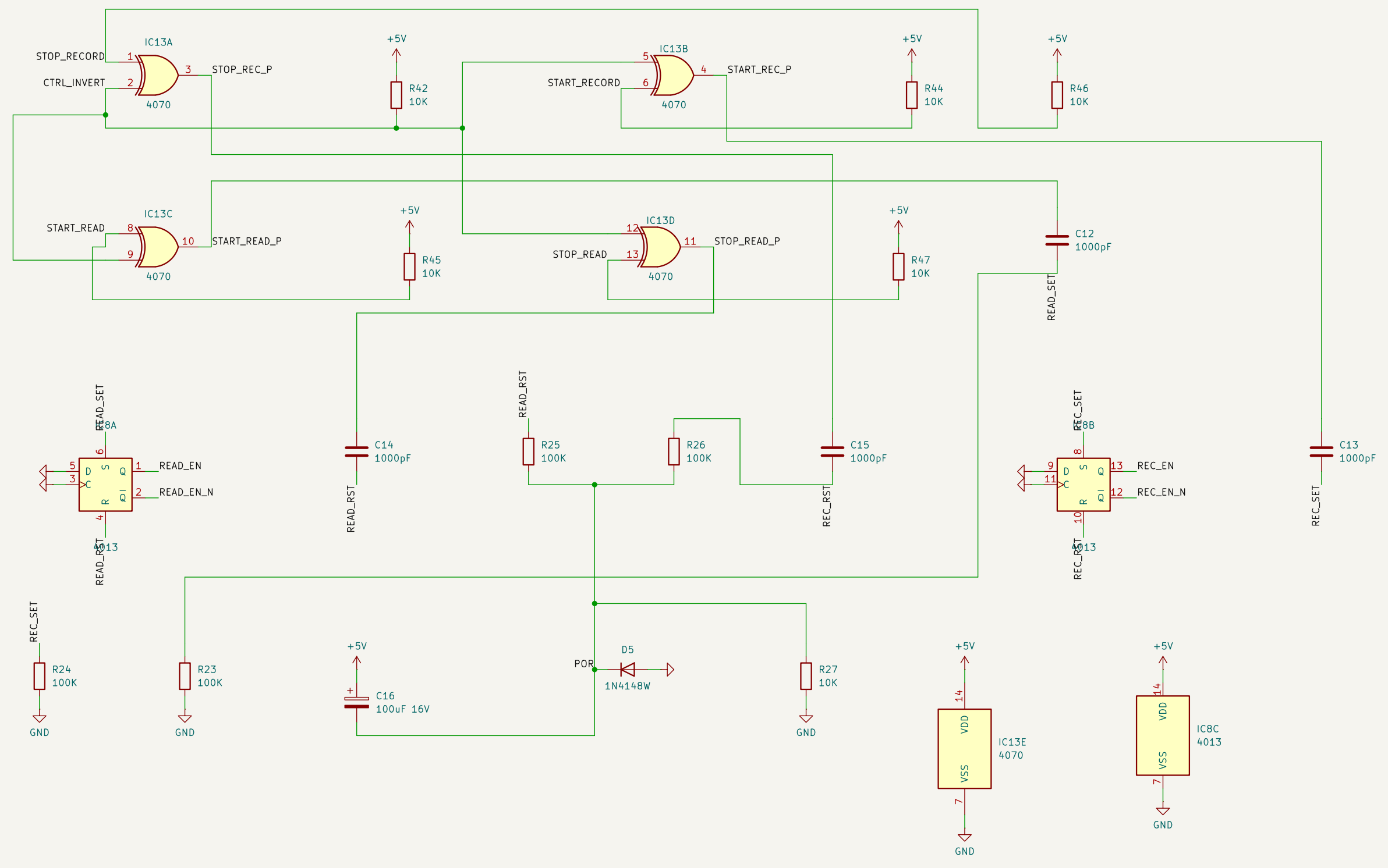
RS−232 receivers / drivers, clock clamps


IC15C
TERM_ECHO
TERM_TX_SEL
TERM_RS232_IN
1488
C19
470pF
GND
IC12A
CPU_RS232_OUT
CPU_DATA
1489
IC12B
TERM_RS232_OUT
TERM_DATA
1489
IC12C
1489
IC12D
1489
+5V
IC12E
1489
GND
IC15A
1488
IC15B
1488
IC15D
CPU_RX
CPU_RS232_IN
1488
C20
470pF
GND
+13V
IC15E
1488
−13V
GND
D8
CPU_CLK_OUT
1N4148W
D9
1N4148W
D10
TERM_CLK_OUT
1N4148W
D11
1N4148W

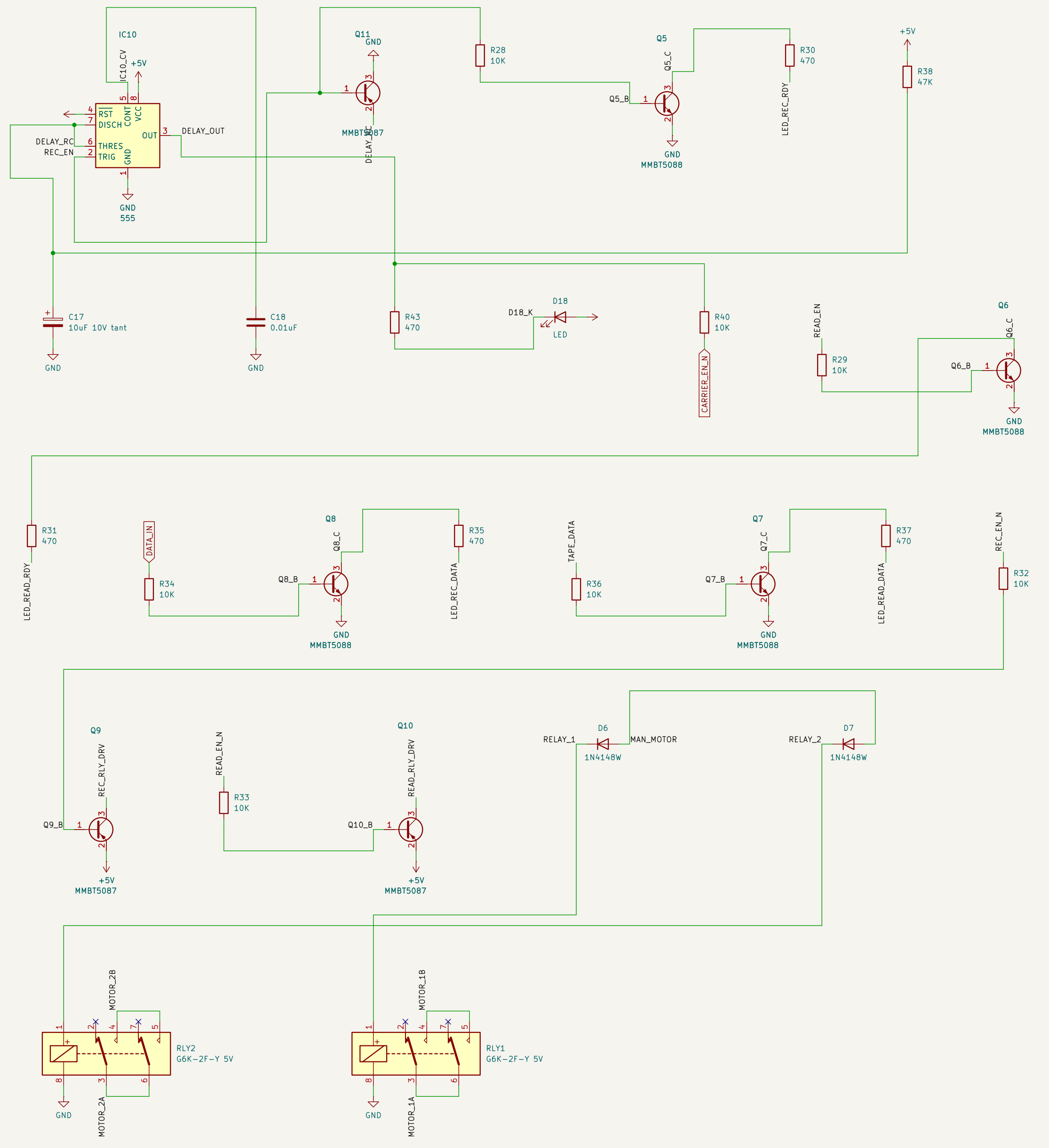
Local/Remote steering and data/clock selectors


IC6
+5V
CPU_DATA
CPU_CLK_OUT
CLOCK_OUT
TERM_CLK_OUT
CPU_CLK_IN
CLOCK_IN
GND
4053
IC14
+5V
TERM_TX_SEL
TERM_ECHO
TERM_CLK_IN
GND
4053
IC11B
LOCAL_REMOTE
SEL_CPU_CLK
4001
IC11A
REMOTE_N
READ_ACTIVE_N
SEL_TERM_CLK
4001
IC7E
4049
+5V
R41
10K
IC11D
CPU_RX_N
4001
IC11C
TERM_DATA_N
TAPE_DATA
ECHO_N
4001
IC7B
TAPE_DATA_N
4049
IC9B
DATA_OUT
CARRIER_DETECT
4023
IC9A
READ_EN
CARRIER_DETECT
4023
IC7D
TERM_DATA
4049
IC7A
CPU_RX
4049
IC7F
DATA_SEL
DATA_SEL_N
4049
IC9C
REC_EN
4023
IC7C
REC_DATA_N
DATA_IN
4049
+5V
IC11E
4001
GND
+5V
IC7G
4049
GND
+5V
IC9D
4023
GND

Record / read control latches and power−on reset


IC13A
STOP_RECORD
CTRL_INVERT
STOP_REC_P
4070
+5V
R42
10K
IC13B
START_RECORD
START_REC_P
4070
+5V
R44
10K
+5V
R46
10K
IC13C
START_READ
START_READ_P
4070
+5V
R45
10K
IC13D
STOP_READ
STOP_READ_P
4070
+5V
R47
10K
C12
1000pF
READ_SET
IC8A
READ_EN
READ_EN_N
4013
READ_RST
C14
1000pF
READ_RST
R25
100K
R26
100K
C15
1000pF
REC_RST
IC8B
REC_SET
REC_EN
REC_EN_N
4013
REC_RST
C13
1000pF
REC_SET
R24
100K
GND
R23
100K
GND
+5V
C16
100uF 16V
POR
D5
1N4148W
R27
10K
GND
+5V
IC13E
4070
GND
+5V
IC8C
4013
GND

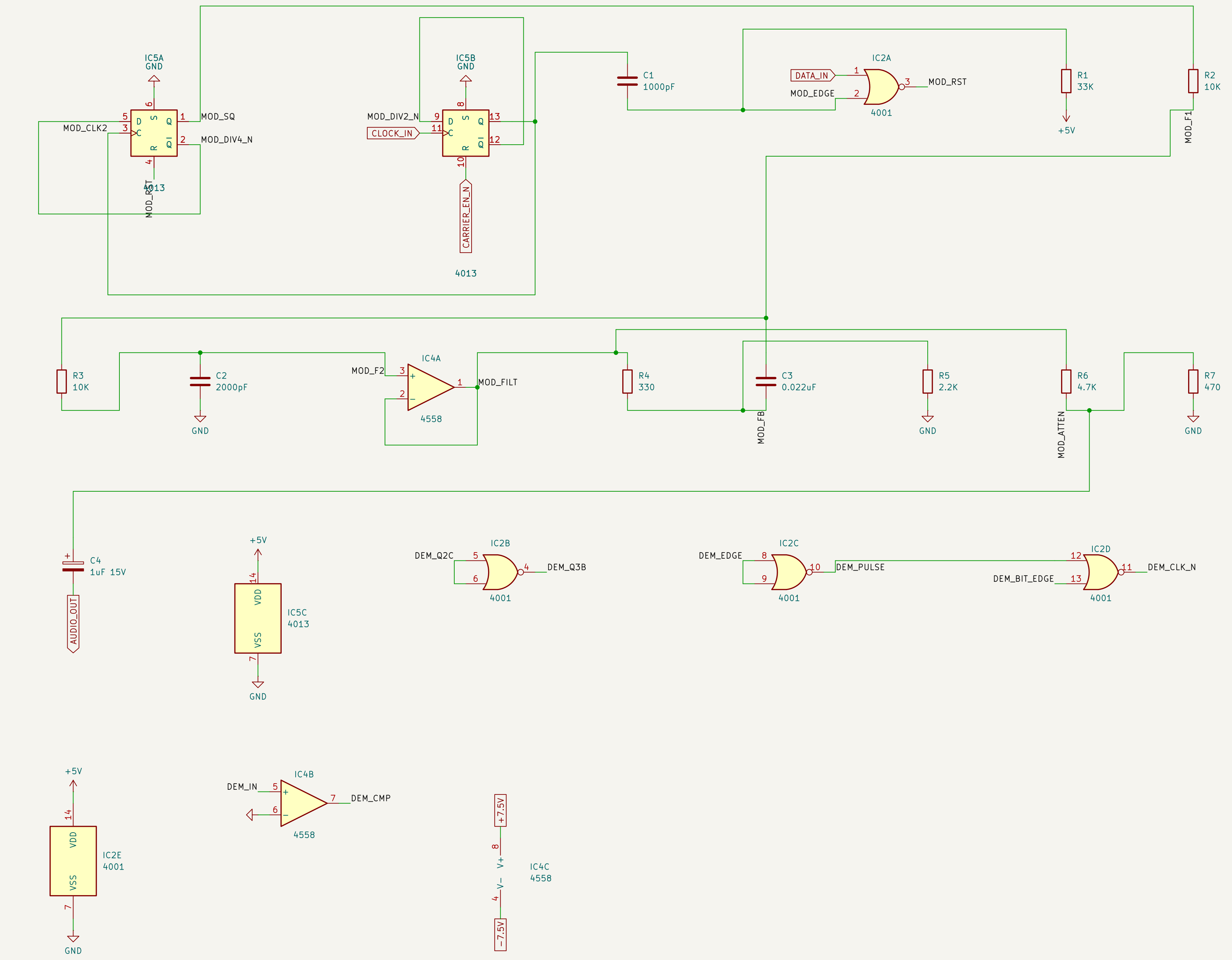
LED drivers, relay drivers, carrier−enable delay


IC10
IC10_CV
+5V
DELAY_RC
REC_EN
DELAY_OUT
GND
555
Q11
GND
MMBT5087
DELAY_C
R28
10K
Q5
Q5_C
Q5_B
GND
MMBT5088
R30
470
LED_REC_RDY
+5V
R38
47K
C17
10uF 10V tant
GND
C18
0.01uF
GND
R43
470
D18
D18_K
LED
R40
10K
CARRIER_EN_N
READ_EN
R29
10K
Q6
Q6_C
Q6_B
GND
MMBT5088
R31
470
LED_READ_RDY
DATA_IN
R34
10K
Q8
Q8_C
Q8_B
GND
MMBT5088
R35
470
LED_REC_DATA
TAPE_DATA
R36
10K
Q7
Q7_C
Q7_B
GND
MMBT5088
R37
470
LED_READ_DATA
REC_EN_N
R32
10K
Q9
REC_RLY_DRV
Q9_B
+5V
MMBT5087
READ_EN_N
R33
10K
Q10
READ_RLY_DRV
Q10_B
+5V
MMBT5087
RELAY_1
D6
1N4148W
MAN_MOTOR
RELAY_2
D7
1N4148W
MOTOR_2B
RLY2
G6K−2F−Y 5V
GND
MOTOR_2A
MOTOR_1B
RLY1
G6K−2F−Y 5V
GND
MOTOR_1A

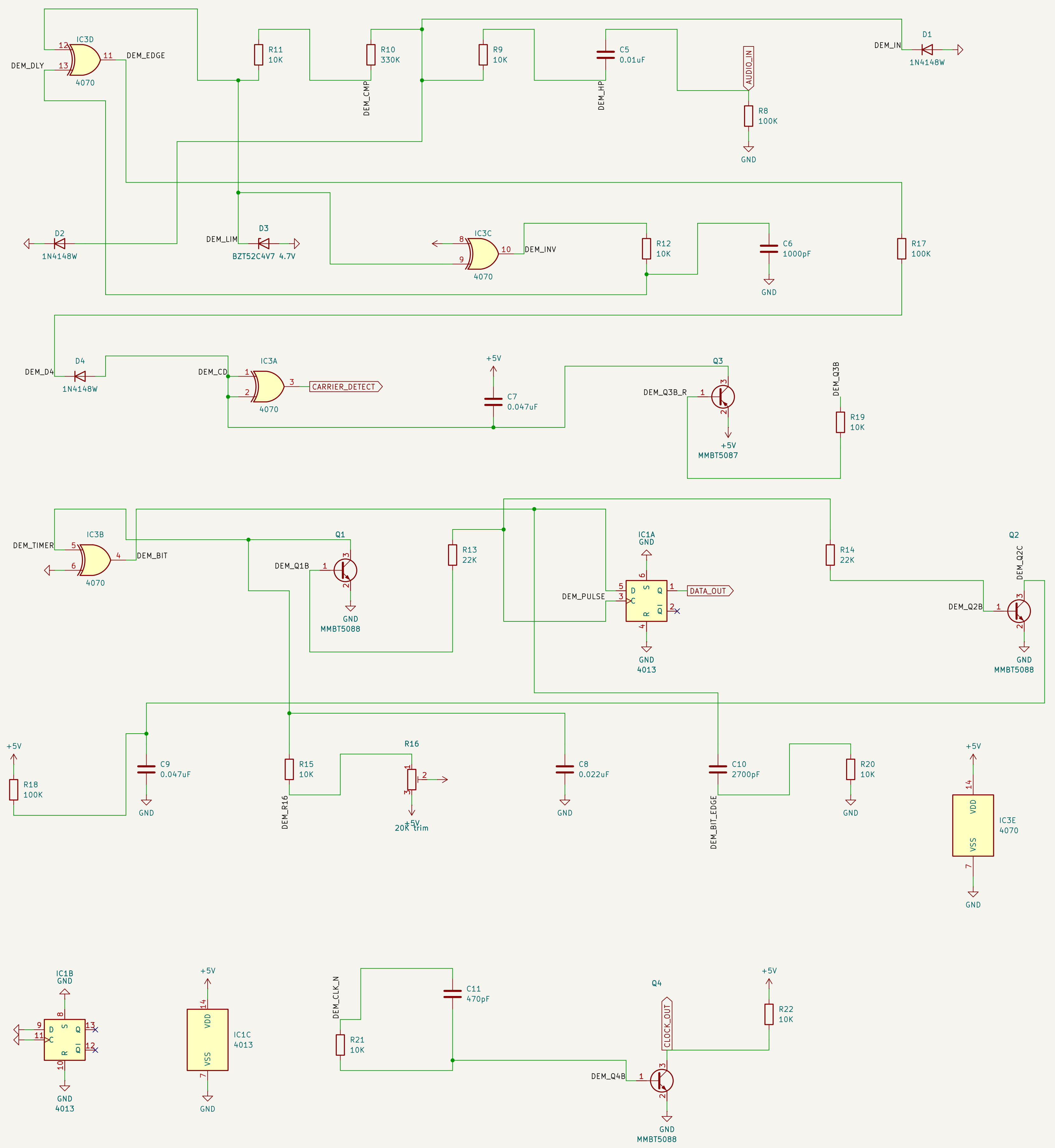
Generated by kicad/tools/gen.py − edit netlist.py and regenerate, or edit in KiCad
Source: docs/ac30_schematica.jpg, docs/ac30_mod_demod.pdf, docs/ac30_PS.pdf, docs/AC30−BOM.m
SWTPC AC−30 reproduction
Sheet: /
File: AC30_SMT.kicad_sch
Title: SWTPC AC−30 Cassette Interface − Switching (SMT)
Size: A0
Date: 2026−10−08
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/3

Generated from docs/ac30_mod_demod.pdf (SWTPC AC–30 Cassette Tape Modulator/Demodulator
Schematic). Each block is wired; net labels join the blocks and sheets (identical labels are connected).
Transcription notes / assumptions:
 – IC4 (4558) supply pins are not drawn. The assembly manual says IC4 runs from the +/–7.5 V
   zener rails (D12/D13 on the power supply sheet).
 – R16's far end has a terminal dot with no label; taken as +5V (Q1/C8 timer pull–up).
 – IC1B is unused and not drawn: its inputs are tied to GND, its outputs left open.
 – The manual's text calls the pulse inverter IC2A; the schematic (followed here) uses IC2C.
Power input: the supply (sheet 3) is built as a separate unit. Its six DC outputs come onto
the board on J12, a 6–pin KK–396 header (1 GND, 2 +5V, 3 +13V, 4 –13V, 5 +7.5V, 6 –7.5V). C23–C25 are +5 V bypass capacitors on the board
(drawn at the regulator on docs/ac30_PS.pdf; next to IC1, IC7 and IC13 on the original artwork).

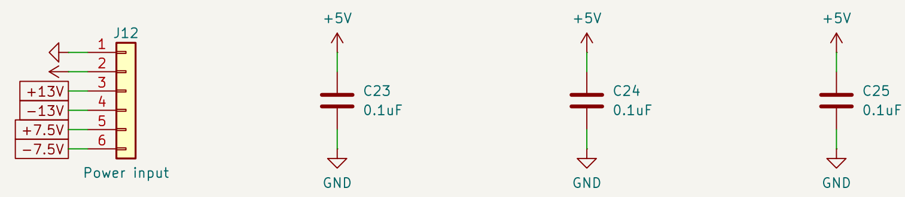
Modulator: CLOCK IN /2 (IC5B), /1 or /2 by DATA IN (IC5A), 2–pole filter IC4A


IC5A
GND
MOD_CLK2
MOD_SQ
MOD_DIV4_N
MOD_RST
4013
IC5B
GND
MOD_DIV2_N
CLOCK_IN
CARRIER_EN_N
4013
C1
1000pF
IC2A
DATA_IN
MOD_EDGE
MOD_RST
4001
R1
33K
+5V
R2
10K
MOD_F1
R3
10K
C2
2000pF
GND
IC4A
MOD_F2
MOD_FILT
4558
R4
330
C3
0.022uF
MOD_FB
R5
2.2K
GND
R6
4.7K
MOD_ATTEN
R7
470
GND
C4
1uF 15V
AUDIO_OUT
+5V
VDD
VSS
IC5C
4013
GND
IC2B
DEM_Q2C
DEM_Q3B
4001
IC2C
DEM_EDGE
DEM_PULSE
4001
IC2D
DEM_BIT_EDGE
DEM_CLK_N
4001
+5V
VDD
VSS
IC2E
4001
GND
IC4B
DEM_IN
DEM_CMP
4558
+7.5V
V+
V–
IC4C
4558
–7.5V

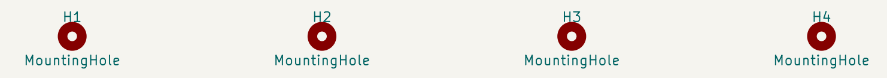
Demodulator: comparator IC4B, edge pulses IC3C/D, carrier detect, data and 16X clock out


IC3D
DEM_DLY
DEM_EDGE
4070
R11
10K
R10
330K
DEM_CMP
R9
10K
C5
0.01uF
DEM_HP
AUDIO_IN
R8
100K
GND
D1
DEM_IN
1N4148W
D2
1N4148W
DEM_LIM
D3
BZT52C4V7 4.7V
IC3C
DEM_INV
4070
R12
10K
C6
1000pF
GND
R17
100K
DEM_D4
D4
1N4148W
DEM_CD
IC3A
CARRIER_DETECT
4070
+5V
C7
0.047uF
Q3
DEM_Q3B_R
+5V
MMBT5087
DEM_Q3B
R19
10K
IC3B
DEM_TIMER
DEM_BIT
4070
Q1
DEM_Q1B
GND
MMBT5088
R13
22K
IC1A
GND
DEM_PULSE
DATA_OUT
GND
4013
R14
22K
Q2
DEM_Q2C
DEM_Q2B
GND
MMBT5088
+5V
R18
100K
C9
0.047uF
GND
R15
10K
DEM_R16
R16
+5V
20K trim
C8
0.022uF
GND
C10
2700pF
DEM_BIT_EDGE
R20
10K
GND
+5V
VDD
VSS
IC3E
4070
GND
IC1B
GND
GND
4013
+5V
VDD
VSS
IC1C
4013
GND
DEM_CLK_N
C11
470pF
R21
10K
Q4
CLOCK_OUT
+5V
R22
10K
DEM_Q4B
GND
MMBT5088

Power input connector (from the external supply) and +5 V bypass


J12
+13V
–13V
+7.5V
–7.5V
Power input
+5V
C23
0.1uF
GND
+5V
C24
0.1uF
GND
+5V
C25
0.1uF
GND

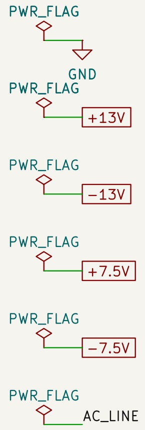
Mounting holes


H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole

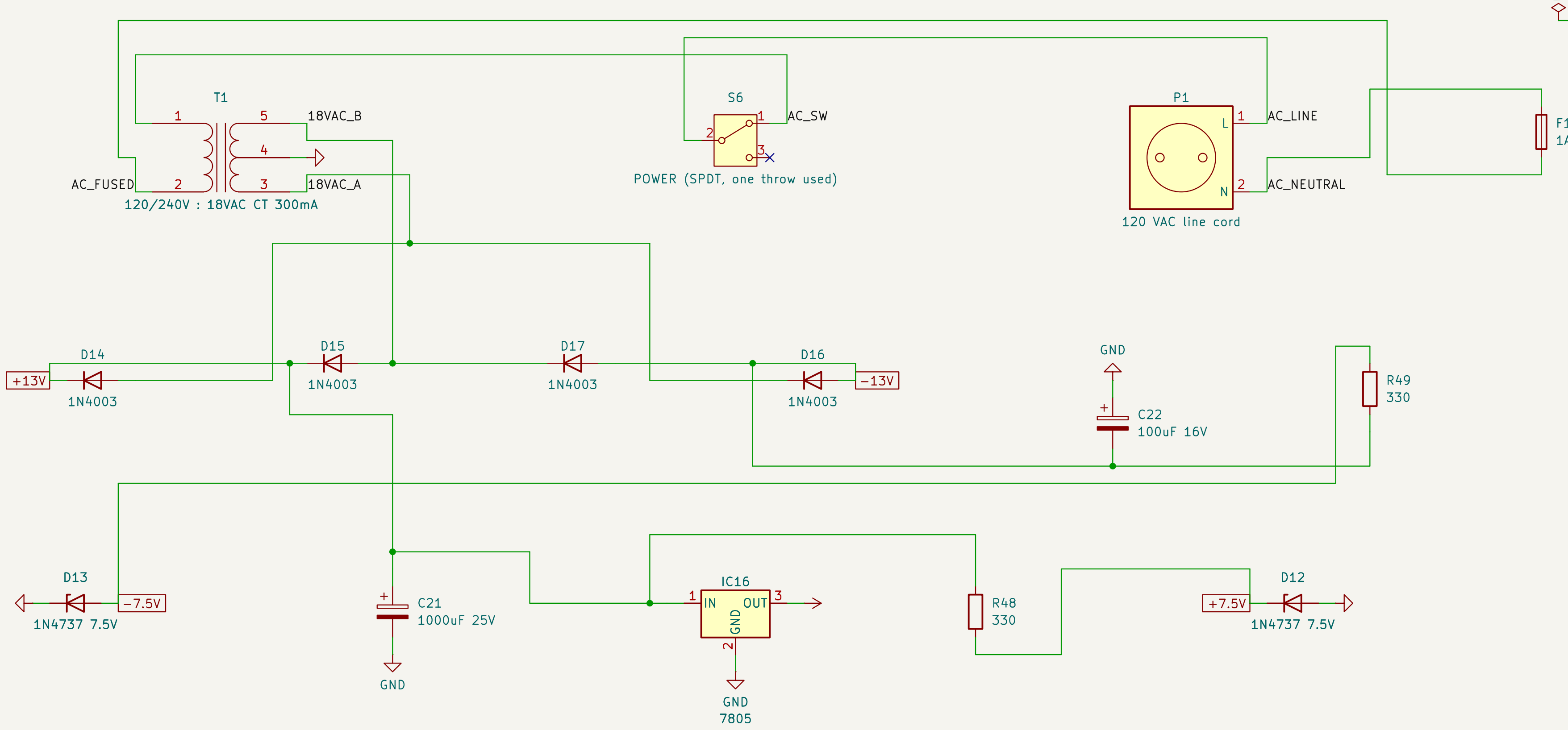
Source: docs/ac30_mod_demod.pdf; power input J12 for the external supply
SWTPC AC–30 reproduction
Sheet: /Mod/Demod & power input/
File: AC30_SMT_moddemod_power.kicad_sch
Title: SWTPC AC–30 – Modulator/Demodulator & power input
Size: A1
Date: 2026–10–08
Rev: A
KiCad E.D.A. 10.0.7
Id: 2/3

Generated from docs/ac30_PS.pdf (SWTPC AC−30 Audio Cassette Power Supply Schematic).
Built as an EXTERNAL unit: these parts have no footprints and are not on the PC board. Wire the six outputs to the board's power input J12 (KK−396: 1 GND, 2 +5V, 3 +13V, 4 −13V, 5 +7.5V, 6 −7.5V).
Transcription notes / corrections:

- The drawing joins the IC16 GND / C23−C25 return line to the bridge's negative corner (−13 V) through two crossings. That would put the 7805's ground at −13 V, so IC16 GND is taken as GND.
- T1's two 120 V primaries are drawn in parallel (120 VAC); drawn here as one primary. The 18 VAC secondary is centre−tapped, tap to GND: the bridge gives +13 V (C21) and −13 V (C22).
- D14−D17 numbering within the bridge is not shown on the drawing; assigned here.
- S6 is an SPDT toggle (BOM) used as an on/off switch; one throw is unused.
- IC15 (1488) supply pins are not drawn on the switching schematic; taken as +/−13 V, the only load for those outputs. IC4 (4558) uses +/−7.5 V (assembly manual).


PWR_FLAG
GND
PWR_FLAG
+13V
PWR_FLAG
−13V
PWR_FLAG
+7.5V
PWR_FLAG
−7.5V
PWR_FLAG
AC_LINE

Power supply (external): 120 VAC −> T1 −> bridge D14−D17 −> +/−13 V, zeners +/−7.5 V, 7805 +5 V


T1
1
5
18VAC_B
4
AC_FUSED
2
3
18VAC_A
120/240V : 18VAC CT 300mA
S6
2
1
AC_SW
3
POWER (SPDT, one throw used)
P1
L
1
AC_LINE
N
2
AC_NEUTRAL
120 VAC line cord
F1
1A
D14
+13V
1N4003
D15
1N4003
D17
1N4003
D16
−13V
1N4003
GND
C22
100uF 16V
R49
330
D13
−7.5V
1N4737 7.5V
C21
1000uF 25V
GND
IC16
1
IN
OUT
3
GND
2
GND
7805
R48
330
D12
+7.5V
1N4737 7.5V

Source: docs/ac30_PS.pdf; parts have no footprints − built as a separate unit
SWTPC AC−30 reproduction
Sheet: /Power supply (external)/
File: AC30_SMT_power_supply.kicad_sch
Title: SWTPC AC−30 − Power supply (external, not on PCB)
Size: A3
Date: 2026−10−08
Rev: A
KiCad E.D.A. 10.0.7
Id: 3/3